Automated Verification and the Enforcement of Eligibility Rules

Evidence from Unemployment Insurance

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Motivation: Eligibility Rules vs. Enforcement

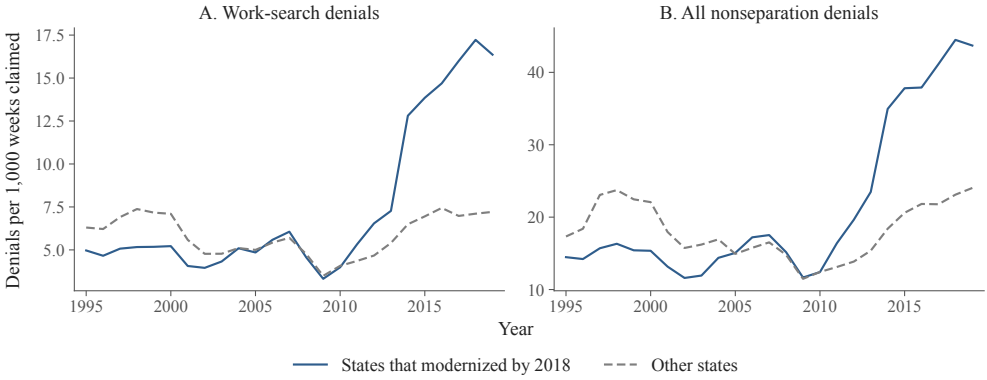
- Access to social insurance depends not only on **eligibility rules**, but also on **how those rules are enforced** (Hacker, 2004; Herd and Moynihan, 2018).

- Governments increasingly use **digital systems and automated verification** to monitor continued eligibility.

- This creates a fundamental trade-off:
  - **Better targeting:** Detect noncompliance and reduce improper payments.
  - **Administrative exclusion:** Increase compliance costs and potentially deny benefits to otherwise eligible claimants.

- In unemployment insurance, claimants must certify every week that they are able, available, and **searching for work**; states now check these certifications through new online benefit systems.

A Growing Gap in UI Eligibility Denials



After moving broadly together before 2010, work-search denial rates diverged sharply: by 2019, 16.4 denials per 1,000 weeks claimed in modernizing states versus 7.2 in other states.

Notes: Claims-weighted denial rates, 1995–2019; 22 states modernized by 2018 vs. other states. Source: DOL ETA 207 and 5159.

Research Question

Central Question

Does automating the verification of an **unchanged** eligibility rule change how often it is enforced?

Two competing mechanisms:

Better enforcement

- More effective identification of genuine noncompliance.
- Fewer improper benefit payments.

Administrative exclusion

- Greater reporting and documentation burdens.
- Potential benefit denials among otherwise eligible claimants.

Both mechanisms can increase denials, but imply different consequences for social insurance protection.

This Paper

- **Setting:** Staggered modernization of state unemployment insurance systems, with 14 first-wave launches during 2013–2018.
- **Data:** Quarterly administrative records on UI eligibility determinations and denials across U.S. states, 2010–2019.
- **Identification:** Compare changes in treated states around system go-live dates with changes in never-modernizing states using stacked difference-in-differences.
- **Mechanisms:** Separate changes in the frequency of eligibility determinations from changes in the probability of denial conditional on a determination.

Preview of Results

1. **Modernization substantially increases work-search denials.**
   - Denials rise by **0.74 log points (≈110%)** following system launch.

2. **More eligibility issues are formally adjudicated, but the share denied remains unchanged.**
   - Formal determinations rise by approximately **0.74 log points**.
   - No statistically significant change in the denial share conditional on determination.

3. **The increase is robust to alternative specifications.**
   - Similar results using Callaway–Sant'Anna and alternative control specifications.
   - Estimates remain substantial when excluding states with contemporaneous work-search law changes.

Related Literature and Contributions

- **Administrative burdens and access to social insurance**
  (Herd and Moynihan, 2018; Deshpande and Li, 2019; Finkelstein and Notowidigdo, 2019; Homonoff and Somerville, 2021)

  → **This paper:** Studies how administrative modernization changes the enforcement of continuing eligibility requirements.

- **Technology and public program administration**
  (Ebenstein and Stange, 2010; Wu and Meyer, 2023; The Century Foundation et al., 2020)

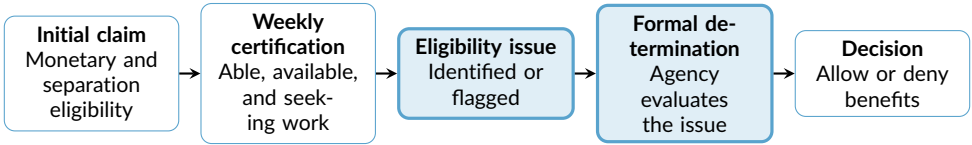
  → **This paper:** Exploits staggered state system launches to estimate changes in eligibility determinations and denials.

- **Unemployment insurance and work-search enforcement**
  (Ashenfelter et al., 2005; McQuillan and Moore, 2025)

  → **This paper:** Shows that modernization is followed by more formal determinations and denials, without a detectable increase in the conditional denial rate.

Background

How Unemployment Insurance Eligibility Is Enforced



- **Continuing eligibility:** Claimants must satisfy requirements such as being available for work and conducting required job searches.
- **Nonseparation denials:** Denials arising from eligibility issues other than the circumstances of job loss; work-search-related denials are one component.
- **Where modernization may matter:** How eligibility information is collected, issues are identified, and cases enter formal adjudication.

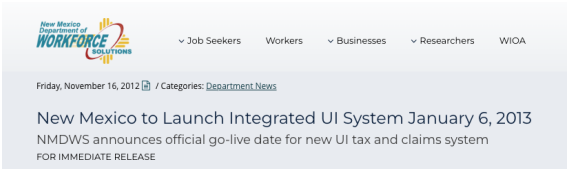
Modernization can change how often eligibility issues are formally examined, even without a change in statutory requirements.

What Does UI System Modernization Mean?

System Go-Live

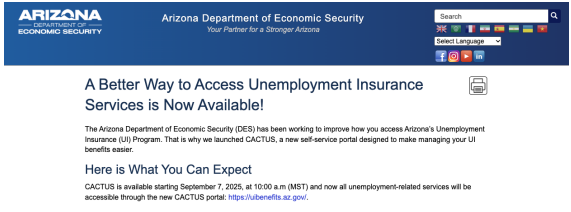
The date a state launches a replacement unemployment insurance benefits system.

New Mexico: 2013 launch



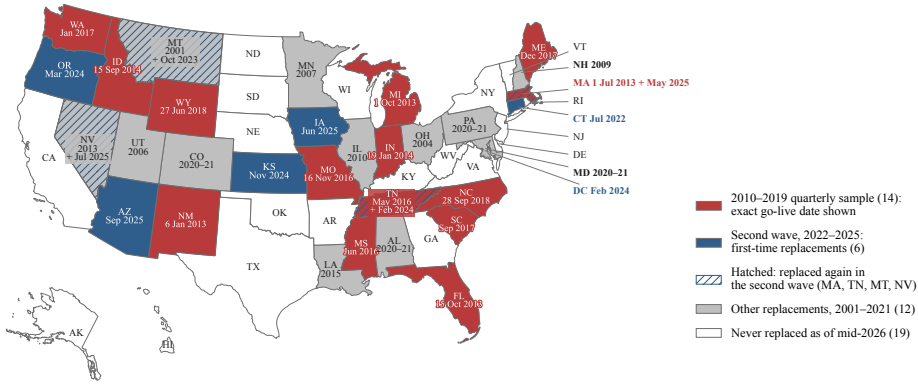
New tax and claims system launched **January 6, 2013**.

Arizona: 2025 launch



CACTUS portal launched **September 7, 2025**.

UI Modernization Occurred at Different Times



The main analysis exploits 14 state system launches during 2013–2018, using quarterly data from 2010–2019.

Notes: Red = main-analysis launches; blue = first replacements in 2022–2025; grey = other replacements in 2001–2021; hatching = repeat replacements in 2023–2025; white = no replacement by mid-2026. Sources: TCF (2020), state announcements, and legislative audits.

Data and Research Design

Data: Measuring UI Eligibility Enforcement

- **DOL administrative records**
  - **ETA 207:** Formal eligibility determinations and denials, by issue.
  - **ETA 5159:** Weeks claimed, weeks paid, and initial claims.

- **System modernization dates**
  - State-specific benefits-system go-live dates from administrative announcements and public records.
  - **14 launches during 2013–2018** in the main estimation sample.

- **Additional data**
  - Work-search laws, unemployment rates, claimant characteristics, and DOL audits.

Main Outcome

$$Y_{st} = \log\left(\frac{\text{Work-search denials}_{st}}{\text{Weeks claimed}_{st}} \times 1{,}000\right)$$

State-quarter panel, 2010Q1–2019Q4; seasonally adjusted. Source: DOL ETA 207 and ETA 5159.

All data sources Variables

Research Design: Stacked Difference-in-Differences

- **Treatment:** State benefits-system go-live, staggered across 14 launches (2013–2018).
- **Comparison:** For each launch cohort, compare treated states with states that never modernized during the relevant study period.
- **Event window:** Up to 12 quarters before and after launch, within 2010Q1–2019Q4.

$$Y_{sct} = \alpha_{sc} + \lambda_{ct} + \sum_{k \neq -1} \beta_k \mathbf{1}[t - g_c = k] D_{sc} + \varepsilon_{sct}$$

- α_{sc}: Stack $\times$ state fixed effects; λ_{ct}: stack $\times$ calendar-quarter fixed effects.
- Standard errors clustered by state; event time $k = -1$ omitted.
- **Identifying assumption:** Absent modernization, treated states would have followed the same outcome trends as their comparison states.

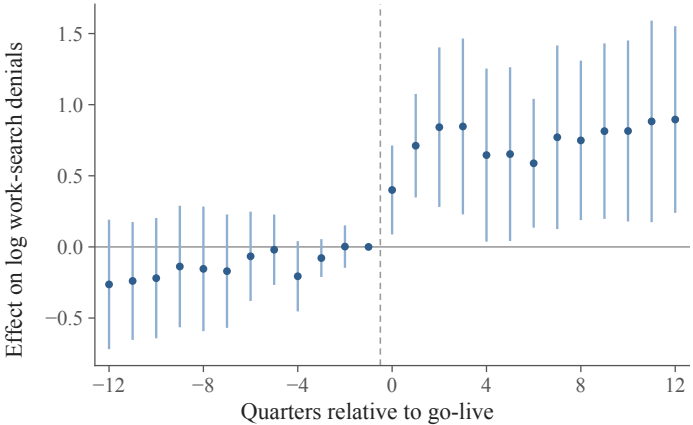
Pre-Launch Characteristics of Treatment and Control States, 2001–2008

	Modernizing states	Comparison states	Difference (SE)
Work-search denials per 1,000 weeks	5.26	6.40	−1.15 (1.55)
Nonseparation denials per 1,000 weeks	17.68	19.09	−1.42 (4.13)
Separation denials per 100 new claims	18.62	24.47	−5.85 (3.73)
First payments per 100 new claims	75.52	79.98	−4.47 (3.44)
Unemployment rate (%)	5.29	4.83	0.46 (0.29)
Average weekly benefit ($)	232.0	236.2	−4.3 (13.5)
Trust-fund solvency	2.60	2.30	0.30 (0.72)
Change in log work-search denials, 2001–02 to 2007–08	−0.006	0.088	−0.094 (0.169)
States	14	29	

Takeaway: No statistically significant differences in these pre-modernization characteristics.

Main Results

Modernization Increases Work-Search Denials



Average Treatment Effect

$+0.740$

log points

SE = 0.273

$\approx$ 110% **increase**

Pre-Trend Test

$p = 0.71$

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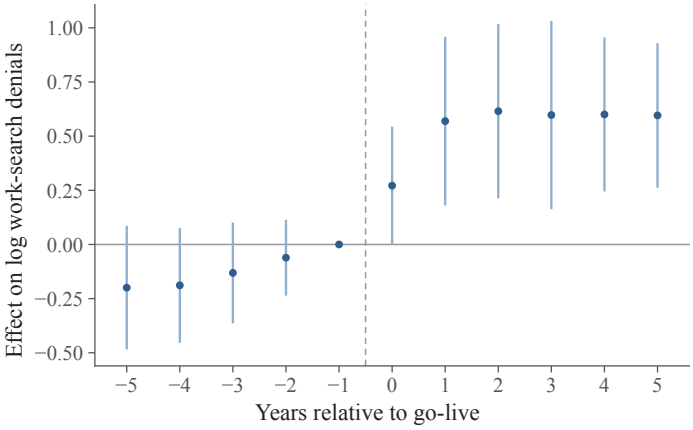
14 launches (2013–2018)

Work-search denials increase substantially following UI system modernization.

Outcome: log denials per 1,000 weeks claimed, seasonally adjusted. Stacked event study, 2010Q1–2019Q4; ATT averages quarters 0–12. SE clustered by state. Pre-launch mean (SD): 11.7 (15.0) denials per 1,000 weeks.

Full Estimates

Annual Data: All 22 Modernizing States



Average Treatment Effect

$+0.542$

log points

SE = 0.173

$\approx$ 72% **increase**

Pre-Trend Test

$p = 0.69$

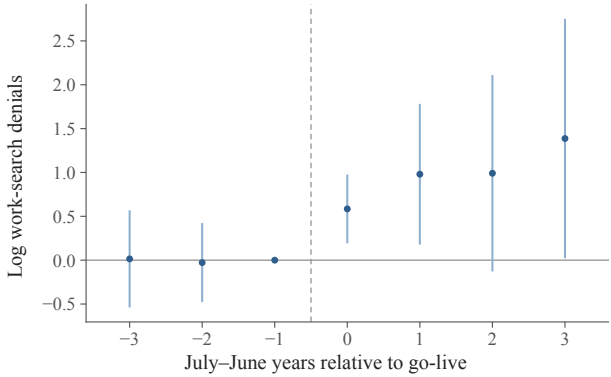
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22 modernizing states

The increase in recorded work-search denials persists when all 22 modernizing states are included in the annual analysis.

Annual stacked event study, 1995–2019, using launch year and a $\pm$5-year window. ATT averages years 0–5. Outcome: log denials per 1,000 weeks claimed. SE clustered by state. Pre-launch mean (SD): 8.0 (10.0) denials per 1,000 weeks.

Validation: Do Denials Rise in the DOL Audit (BAM) Sample?



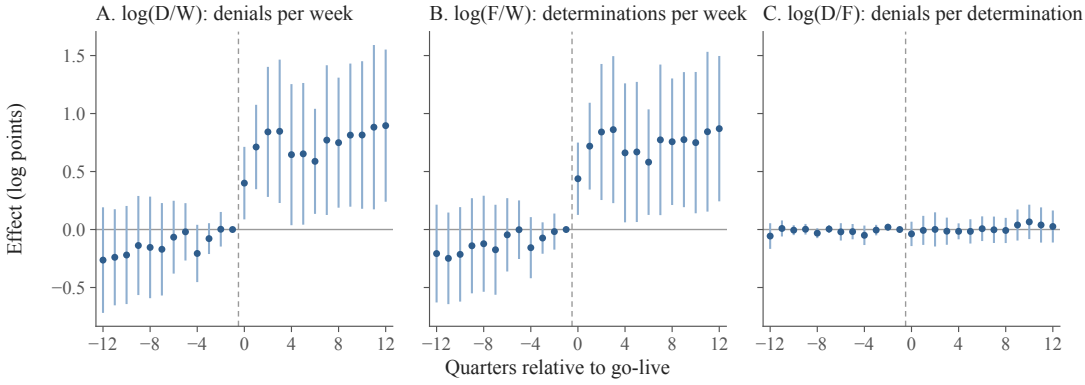
In the states and years covered by the DOL audits, work-search denials rise by 0.99 log points (SE 0.33).

Log work-search denials per 1,000 weeks claimed, state × July–June year, July 2011–June 2019; 9 states, ±3 years. Pre-trend $p = 0.84$.

Audit results ◂ Back

Mechanisms

Where Does the Increase in Denials Come From?

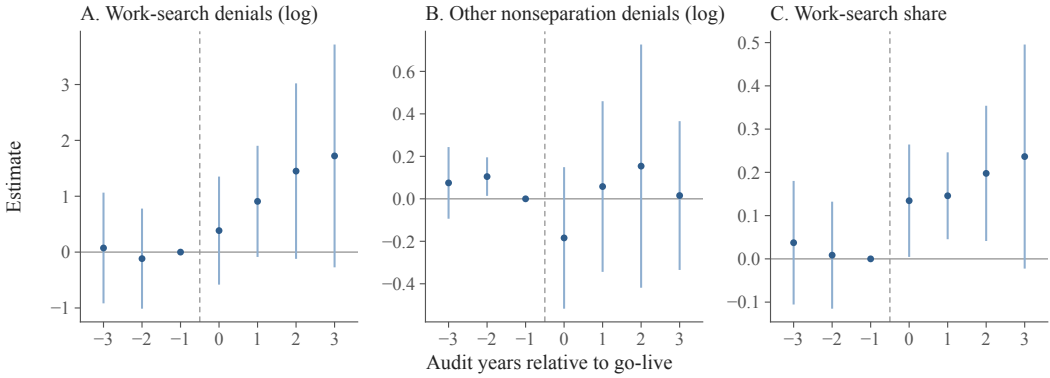


Modernization increases formal eligibility determinations, with no detectable change in the share of determinations resulting in denial.

D = denials; F = formal determinations; W = weeks claimed. $\log(D/W) = \log(F/W) + \log(D/F)$. Formal determinations do not directly measure genuine violations or automated flags.

Decomposition Example

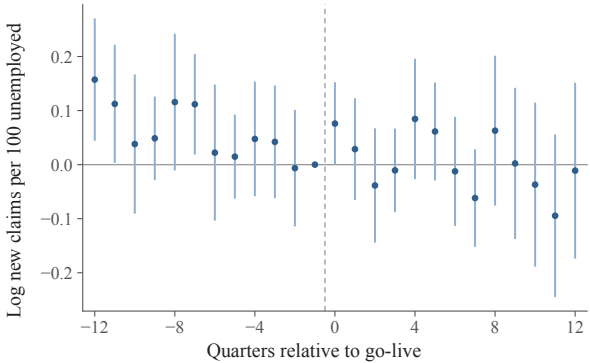
Is the Increase Concentrated in Work-Search Denials?



Work-search-related denials increase, while the estimated effect on other nonseparation denials is close to zero.

DOL BAM sample: work-search +1.12 (SE 0.57); other nonseparation issues +0.01 (SE 0.20). Nine launching states, 2012–2019. Issue classification differs from ETA 207.

Do Applications and Claiming Change Around Modernization?



We find no clear post-launch break in initial claims or weeks claimed, although rejected pre-trend tests limit causal interpretation.

ETA 5159 new initial claims and weeks claimed per unemployed worker (DOL, 2026b; BLS, 2026), log, seasonally adjusted, 2010–2019.

Back to Mechanisms

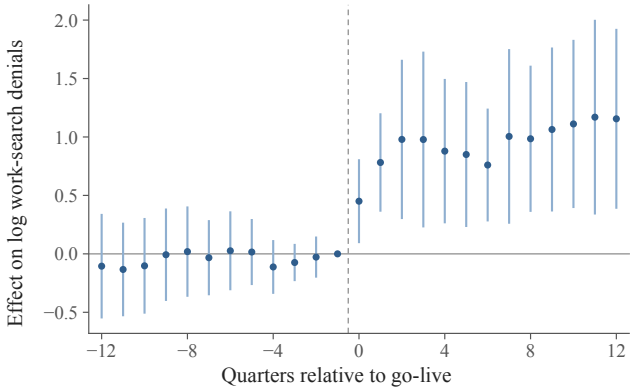
Identification Concerns and Empirical Tests

1. **Concurrent Policy Changes**
   States may tighten work-search requirements around modernization.
   → **Test:** Exclude states that increased required employer contacts near launch.

2. **Fiscal Incentives**
   States may increase enforcement to reduce UI benefit expenditures.
   → **Test:** Examine federally financed UCFE claims processed by the same systems.

3. **Fiscal Pressure**
   States with depleted trust funds may tighten enforcement to cut benefit costs.
   → **Test:** Compare the jump in denials at go-live in low- and high-solvency states.

Concern 1: Are Work-Search Law Changes Driving the Result?



Exclude states that tightened work-search requirements near launch

Sample	Effect (SE)	States
All launches	0.740 (0.273)	14
Exclude MI, MS, MO	0.936 (0.307)	11

Pre-Trend Test

Full sample: $p = 0.71$

Restricted sample: $p = 0.11$

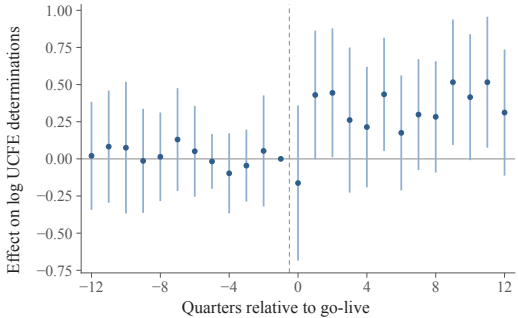
The increase in work-search denials persists after excluding states that tightened their requirements near launch.

Required employer contacts from DOL's *Comparison of State UI Laws*. The exclusion addresses observed changes in required contacts, not all concurrent policy or administrative reforms.

Additional Law Checks

Concern 2: Are Fiscal Incentives Driving the Result?

Federally financed UCFE claims processed through state UI systems



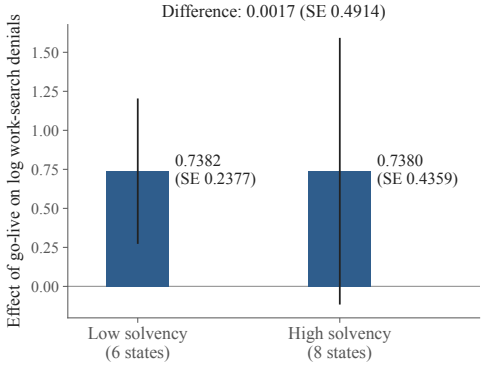
Effect on formal determinations: $+0.32$ log points (SE 0.18)

Formal determinations also increase for federally financed claims, consistent with broader changes in administrative processing rather than an explanation based solely on state benefit financing.

Outcome: log UCFE determinations per 1,000 UCFE weeks claimed. Quarterly, 2010–2019; 14 launches.

Additional Placebos

Concern 3: Is Fiscal Pressure Driving the Result?



Effect of Go-Live by 2007 Solvency

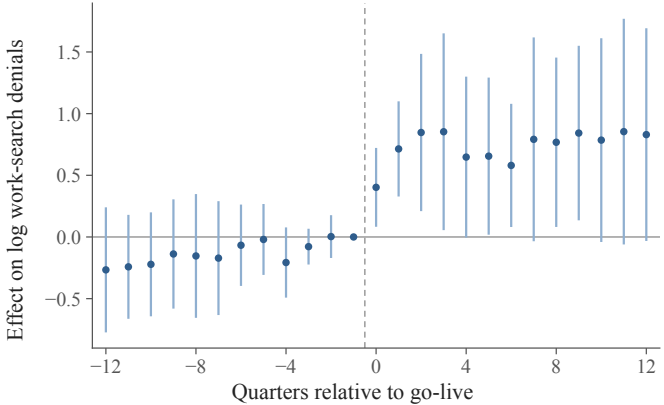
Group	Effect of go-live	% increase in denials
Low solvency IN, MA, MI, MO, NC, SC	0.7382 (0.2377)	+109.22%
High solvency FL, ID, ME, MS, NM, TN, WA, WY	0.7380 (0.4359)	+109.17%
Difference	0.0017 (0.4914)	

Denials roughly double at go-live in states with depleted and with healthy trust funds: low-solvency states do not behave differently.

Solvency: Dec. 2007 trust-fund balance ÷ mean annual benefits, 2005–2007; low = bottom third of all states. Effect: average of event quarters 0–12 from the main stacked event study (log work-search denials, 2010Q1–2019Q4), run separately for each group with the same never-modernized controls; % increase $= e^{b} - 1$. Difference: pooled model interacting event time with low solvency. SE clustered by state in parentheses; 95% CIs in the figure. Source: DOL ETA 207, 5159, 2112 (DOL, 2026b).

Robustness

CSDID Estimator



Average effect

Estimator	Effect
Stacked DID	0.74 (0.27)
Callaway–Sant'Anna	0.74 (0.32)

Alternative controls:
Not-yet-treated states
0.72 (SE 0.33)

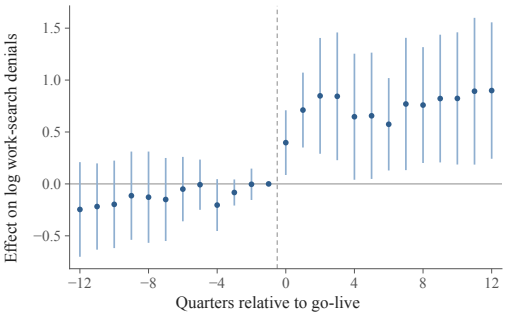
The estimated increase in work-search denials is nearly identical across alternative DID specifications.

14 first-wave launches, 2010–2019. Callaway–Sant'Anna estimates group-time effects with state-clustered multiplier bootstrap. Comparison groups as indicated.

Estimator Details Additional Outcomes

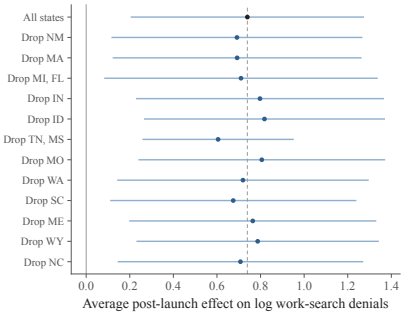
Is the Result Driven by Labor Markets or One Launch Cohort?

Controlling for Unemployment



Effect: **0.742** (SE 0.272)

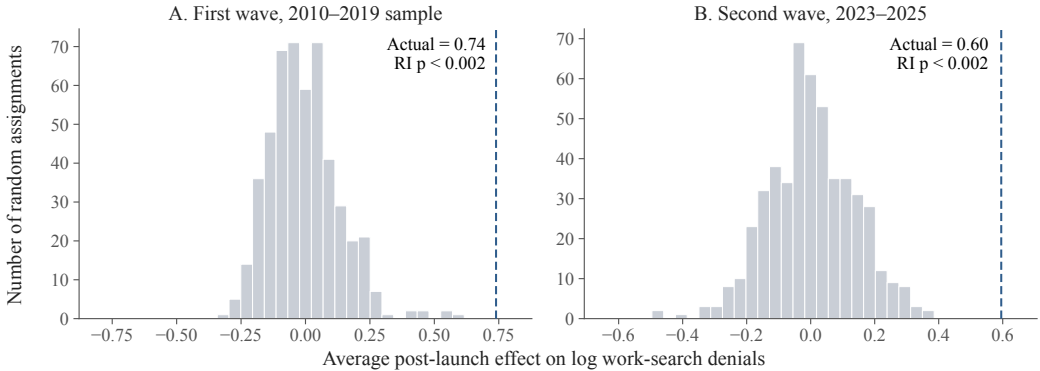
Leave-One-Cohort-Out



Estimates range from **0.61 to 0.82**

The estimated increase remains substantial after controlling for unemployment and excluding individual launch cohorts.

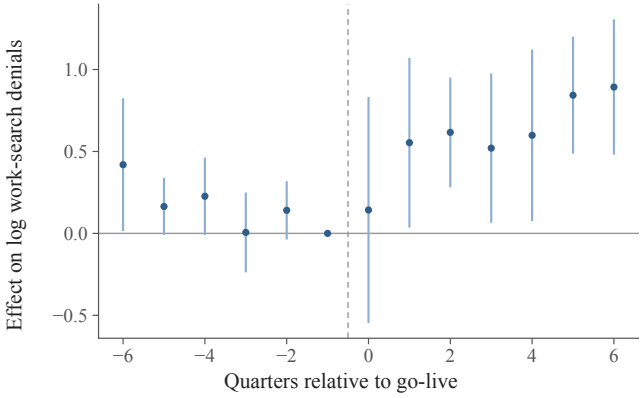
Few Treated States? Randomization Inference



500 random reassignments of the launch quarters to states: none is as large as the actual estimate. RI $p < 0.002$ in both waves.

Later Wave: 2023–2025 Launches

Does the Increase Reappear in the 2023–2025 Wave?



Later-wave estimate

+0.595

log points (SE 0.201)

≈ 81% increase

First-wave estimate

+0.740 (SE 0.273)

Sample

9 launches, 2023–2025

2022Q1–2026Q2

A later wave of UI system launches is also followed by substantially higher recorded work-search denial rates.

Stacked event study; up to 6 quarters around launch. The later-wave estimates require separate assessment of pre-trends and comparison-group validity.

How Large Is the Increase in Recorded Denials?

Measure	Magnitude
Estimated increase in denial rate	$\approx$ 110%
Additional denials per 1,000 weeks claimed	**5.3–12.8**
Equivalent share of weeks claimed	0.5–1.3%
Illustrative aggregate extrapolation	
Additional recorded denials over the first six years	1.27 million
One-week benefit equivalent ($299 per denial)	$380 million

Modernization is associated with a substantial increase in recorded eligibility denials, both in relative and absolute terms.

Conclusion

1. **Modernization changes recorded enforcement.**
   Work-search denial increase by 0.74 log points ($\approx$ 110%) following UI system modernization.

2. **The increase operates through more formal eligibility determinations.**
   Determinations rise substantially, while the conditional denial share shows no detectable change.

3. **The result is robust across several specifications.**
   The increase persists under alternative DID estimators, labor-market controls, and exclusions of selected launch cohorts and states with work-search law changes.

Administrative technology can change how eligibility rules operate in practice, even without a uniform change in statutory rules.

Next Step: Claimant-Level Administrative Records

Plan: Apply for claimant-level administrative records from a state UI agency in the main sample.

What I will request

- Weekly certifications and reported work-search contacts.
- Issues flagged: by the system or by staff.
- Determinations, denials, appeals, and reversals.
- Linked UI wage records: reemployment and earnings.

What it answers

- **Who is flagged?** New vs. continuing claimants; age, education, industry.
- **Targeting or exclusion?** Reversal rates on appeal; denied claimants later found eligible.
- **Consequences:** Weeks lost, reemployment, and earnings after a denial.

Research Design: Callaway–Sant'Anna

- Estimates a separate effect for each launch cohort g and quarter t:

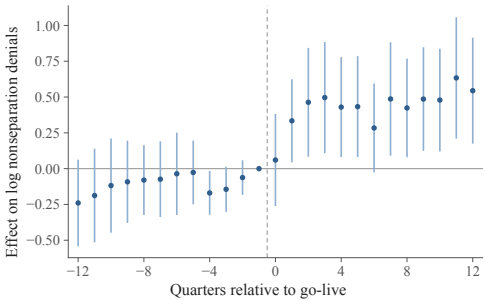
$$ATT(g,t) = E[\,Y_t - Y_{g-1} \mid G = g\,] - E[\,Y_t - Y_{g-1} \mid C\,]$$

- Comparison group C: **never-treated** states, or **not-yet-treated** states (launch later)
- Aggregates $ATT(g,t)$ to event time with explicit weights; no already-treated states used as controls (Callaway and Sant'Anna, 2021)
- Same sample and cohorts as the stacked design; outcome regression, multiplier bootstrap clustered by state

Two estimators that avoid the bias of standard two-way fixed effects with staggered timing.

Weekly Eligibility versus Reason for Job Loss

A. Nonseparation denials

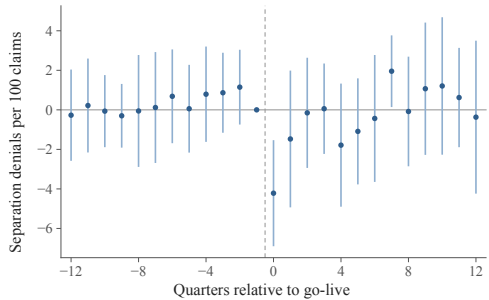


$+0.427^{***}$ (0.160)

"Are you still eligible this week?"

asked every week, checked by the new systems

B. Separation denials



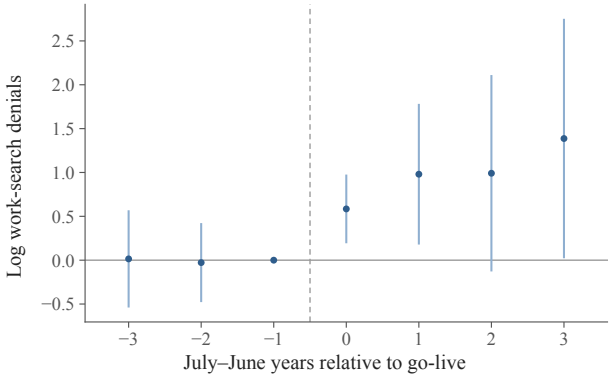
-0.36 (0.98)

"Why did you lose your job?"

asked once, at the initial claim

A: log denials per 1,000 weeks claimed (pre-trend test rejects for this broader category). B: denials per 100 new claims. Quarterly, 14 states, 2010–2019.

Validation: Do Denials Rise in the DOL Audit (BAM) Sample?

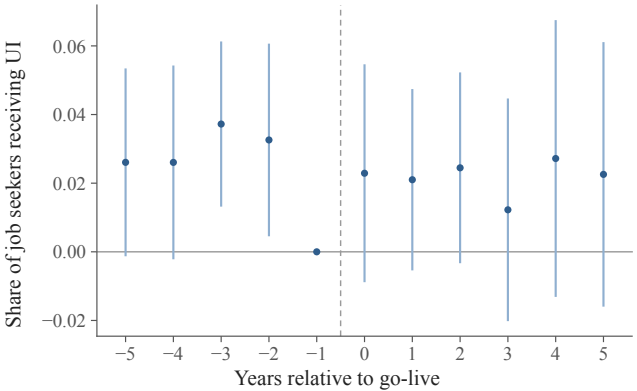


In the states and years covered by the DOL audits, work-search denials rise by 0.99 log points (SE 0.33).

Log work-search denials per 1,000 weeks claimed, state × July–June year, July 2011–June 2019; 9 states, ±3 years. Pre-trend $p = 0.84$.

Audit results ◂ Back

Do Claimants Lose Weeks, or Lose UI Altogether?

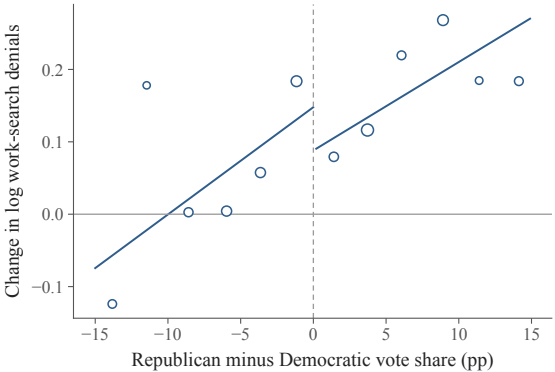


Share receiving any UI	Post − pre (SE)
All job seekers (mean 0.25)	−0.003 (0.010)
High school or less	−0.006 (0.015)
Age 18–29	−0.006 (0.009)
Black	−0.010 (0.020)
Hispanic	0.013 (0.024)

We find no detectable decline in the annual share of job seekers receiving UI; these data cannot establish how additional denials affect benefit duration.

CPS ASEC 2002–2020, adults 18–64 who looked for work, state×year; annual stacked event study, ±5 years.

Do Money or Politics Explain It?

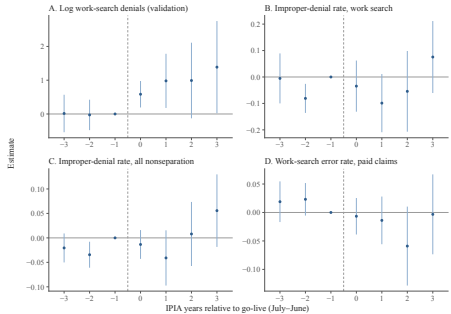


Test	Estimate (SE)
Low 2007 solvency, DiD	0.30 (3.85)
Triple diff., state vs. federal claims	0.007 (0.021)
GOP trifecta 2011, DiD	6.66* (3.46)
Close-election RD, 10 pp	−0.16 (0.22)
Close-election RD, 15 pp	−0.12 (0.15)
Federally paid EUC08 minus regular	0.89 (1.58)

These supplementary tests provide limited evidence that fiscal stress or partisan control alone explains the increase in recorded denials.

Solvency [2007 trust-fund reserves ÷ annual benefits]. DiD with state and year FE and unemployment controls, 2000–2019. RD: governor elections 1990–2015, local linear, outcome = change in log work-search denials in the first term.

Were the Extra Denials Correct?



Improper-denial shares do not rise; paid weeks without job search fall by 3.5 points from 12.8% ($p = 0.12$). Consistent with enforcement, but auditors apply documentation rules, so a searcher who did not document counts as properly denied.

BAM random audits (DOL, 2026a), July 2011–June 2019, 9 states, ± 3 years. Pre-trend tests reject for the improper-denial rates; about 14 audited work-search denials per state-year.

Audit table ◂ Back

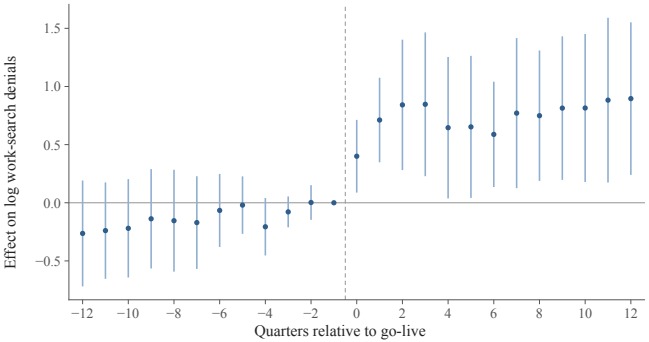
Appendix

Key Variables: Definition and Construction

Variable	Definition	Construction (DOL reports)
Work-search denials (main outcome)	Weeks denied because the claimant was not able, available, or searching for work	1,000× ETA 207 able-and-available denials ÷ weeks claimed; log, seasonally adjusted
Weeks claimed	Weeks for which claimants certified and asked to be paid	ETA 5159 intrastate + interstate-liable weeks claimed
Determinations (detection)	Work-search questions an examiner decided, allowed or denied	1,000× ETA 207 able-and-available determinations ÷ weeks claimed; log
Denial rate (adjudication)	Share of decided questions that end in a denial	ETA 207 denials ÷ determinations
Nonseparation denials	All denials about continuing eligibility (work search, reporting, refusing work, other)	1,000× ETA 207 nonseparation denials ÷ weeks claimed; log
Separation denials (placebo)	Denials over why the job ended, decided at the initial claim	100× ETA 207 separation denials ÷ new initial claims
UCFE denial rate (placebo)	Denial rate on federal employees' claims, same systems, federally paid	ETA 207 UCFE denials ÷ UCFE determinations
Improper-denial rate (BAM)	Share of audited denials that DOL auditors judged legally wrong	BAM denied-claims audits, state × July–June year

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Main Result in Numbers



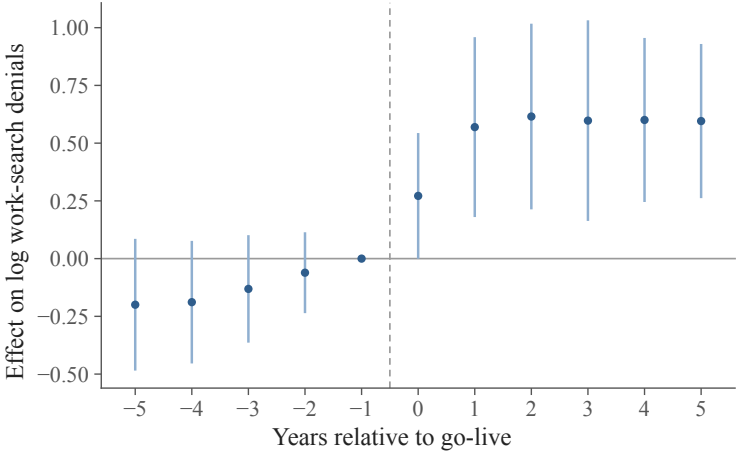
	(1)	(2)
Go-live	0.740***	0.742***
	(0.273)	(0.272)
Pre-trend test, p	0.71	0.70
Unemployment controls	No	Yes
Pre-launch mean (level)	11.7	11.7
Pre-launch SD (level)	15.0	15.0
Treated states	14	14
Observations	8,978	8,420

Denials jump at go-live by 0.74 log points (≈ +110%) and stay up for 3+ years; no pre-trend.

Outcome: log work-search denials per 1,000 weeks claimed, seasonally adjusted. Stacked event study, 14 states launching 2013–2018, 2010Q1–2019Q4. Estimate = average of quarters 0–12; mean and SD of denials per 1,000 weeks in treated states, quarters −12 to −1. SE clustered by state. $^{***}p < 0.01$.

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Annual Data: All 22 First-Wave States, 1995–2019



Log work-search denials per 1,000 weeks

Avg. effect, years 0–5: +0.542*** (0.173)

Pre-trend $p = 0.69$

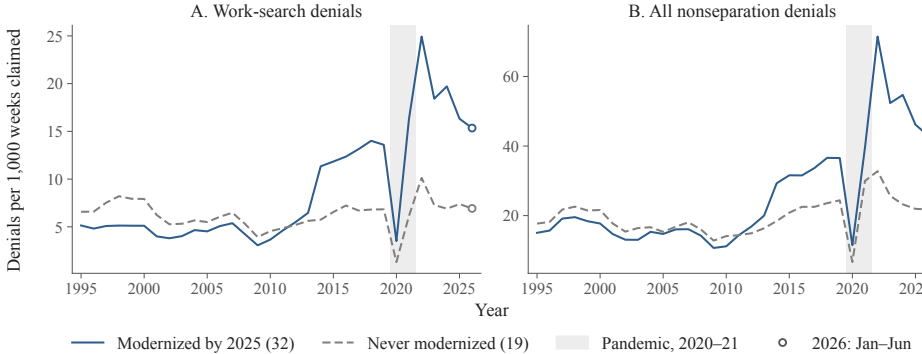
22 states, go-live year, ±5 years

Using every state that replaced its system, including the 5 with only a launch year, the increase is still large and the pre-period is flat.

Annual stacked event study, 1995–2019 (includes the Great Recession years). Smaller than the quarterly estimate because a launch year mixes months before and after go-live.

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Denials, 1995–2026: All Modernized States



Counting every state that replaced its system by 2025, the gap persists: in the first half of 2026, work-search denials are 15.3 per 1,000 weeks vs. 6.9 in states that never modernized.

32 modernized (22 by 2018; AL, CO, MD, PA in 2020–21; AZ, CT, DC, IA, KS, OR in 2022–25) vs. 19 never modernized; membership fixed over time. 2020–21 shaded: weeks claimed rose about sixfold and work-search rules were suspended. 2026 (hollow marker): January–June, the latest half year reported by all states. DOL ETA 207 and 5159 (DOL, 2026b).

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Effect by Wave of Launches

	All dated launches	Unemployment controls
Log work-search denials per 1,000 weeks claimed, 2010–2019		
Go-live	0.740***	0.742***
	(0.273)	(0.272)
Pre-trend test, *p*	0.71	0.70
Pre-launch mean (level)	11.7	11.7
Treated states	14	14

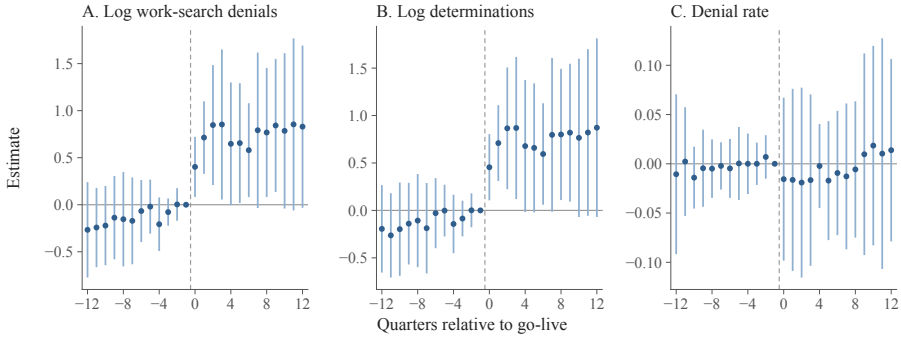
By wave of launches (stacked DiD)	
Full sample: both waves pooled, 2001–2025 (24 states)	0.557*** (0.173)
First wave, 2010–2019 (14 states)	0.740*** (0.273)
Second wave, 2022–2026 (9 states)	0.595*** (0.201)

About 0.6–0.75 log points in every specification and in both waves of launches.

Average of event-quarter coefficients 0–12 (0–6 for the second wave and the pooled sample). Pooled [both waves in one regression; 2020–21 excluded]. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

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Callaway–Sant'Anna: Denials, Determinations, Rate

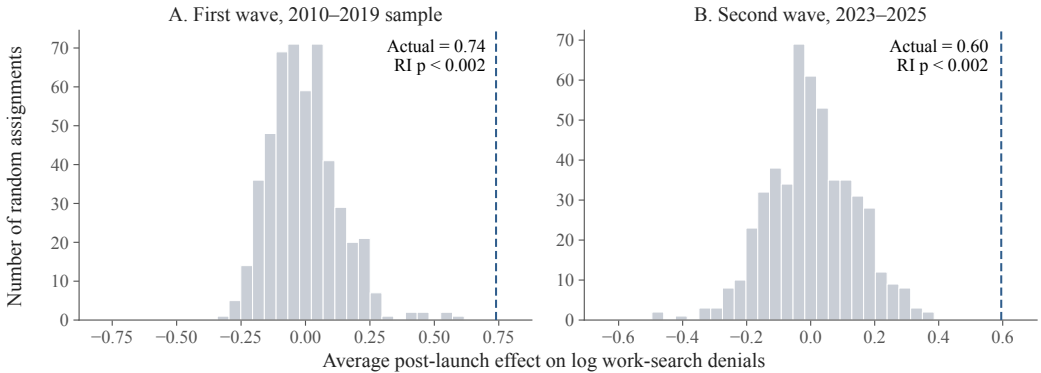


Callaway–Sant'Anna gives 0.74 (SE 0.32) for denials, the same as the stacked estimate.

Callaway and Sant'Anna (2021) [a separate effect for each launch cohort and quarter, averaged with explicit weights], never-modernizing states as controls, 2010–2019, multiplier bootstrap clustered by state. Not-yet-treated controls: 0.72 (0.33).

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Robustness: Few Treated States? Randomization Inference



500 random reassignments of the launch quarters to states: none is as large as the actual estimate. RI $p < 0.002$ in both waves.

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History: How UI Claims Have Been Filed and Checked, 1935–2026

Period	New claim [initial application]	Weekly claim [certification to keep receiving benefits]	Work-search check
1935–1939	In person at the employment office	In person every week	In-person reporting
1940s–early 1990s	In person, one-on-one	Usually by mail	Contacts written on a paper form
1990–1996	Telephone begins (CO; WI 1995; MA, CA 1996)	Touch-tone phone [automated voice system]	Press a button: "I searched"
1997–2006	Phone and internet; offices close; 13% in person by 2006	Phone (47 states), web (29)	Self-reported; little staff contact
2001–2018	Online by default; **22 states replace systems** (17 from 2007)	Online, often only channel	**Each employer contact entered; system flags gaps automatically**
2020–2021	Pandemic surge; identity-verification vendors	Online and phone	Waived or relaxed in many states
2022–2026	Second wave of 10 launches; MI, NY, OH due 2026	Online and mobile	Online entry with automated checks

Earlier technology made filing easier (and did not change take-up); the post-2007 systems are the first to make verification of the same rule stricter.

Wisconsin Blue Book; Social Security Bulletin (1938); O'Leary (2006); Ebenstein and Stange (2010); The Century Foundation et al. (2020); DOL; GAO-23-105478. Practices varied by state; rows describe the typical state.

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Which States Replaced Their Benefits Systems?

Group	States	Number
First wave, 2001–2018	FL, ID, IL, IN, LA, MA, ME, MI, MN, MO, MS, MT, NC, NH, NM, NV, OH, SC, TN, UT, WA, WY	22
Pandemic launches, 2020–21	AL, CO, MD, PA (excluded: work search was suspended)	4
Second wave, 2022–2025	AZ, CT, DC, IA, KS, MA, MT, NV, OR, TN	10
Announced for late 2026	MI, NY, OH	3
Never replaced (mid-2026)	AK, AR, CA, DE, GA, HI, KY, ND, NE, NJ, NY, OK, RI, SD, TX, VA, VT, WI, WV	19

Treatment = go-live of a new benefits system [software that takes claims, runs weekly certification, flags eligibility issues, and pays benefits]**. Identification comes from *when* each state switched, set by multi-year procurement.**

First wave from a 2020 report on modernized states; month or day of go-live found for 17 states (year only for LA, MT, NV, OH, UT). States can appear twice: MA, MT, NV, TN replaced their first-wave systems again in 2023–2025; MI and OH plan a second replacement in late 2026; NY has never replaced its system, and its first replacement is scheduled for late 2026.

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All Data Sources

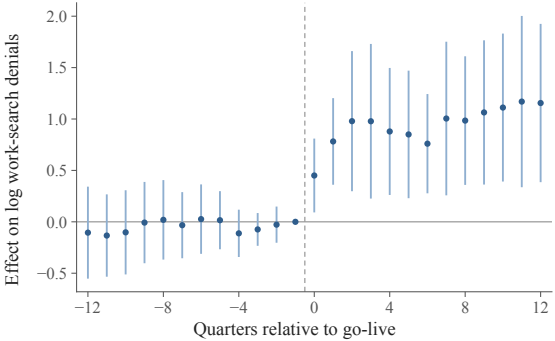
Data set	Unit, years	Used for
ETA 207 nonmonetary determinations	State × quarter, 1995–2026	Denials and determinations by issue (main outcome, mechanism, placebos)
ETA 5159 claims and payments	State × month	Weeks claimed (denominator), weeks paid, first payments
ETA 5130 appeals; ETA 218; ETA 2112	State × month/quarter	Appeals; potential duration; trust-fund solvency
ETA 203 claimant characteristics	State × month, 1995–2019	Composition; heterogeneity
EUC08 versions of ETA 207/5159	State, 2009–2013	Federally financed placebo
BAM audits (denied and paid claims)	State × July–June year, 2012–2020	Accuracy of denials
BLS LAUS; CPI-U	State × month; year	Unemployment controls; deflator
CPS ASEC	Person, 2002–2020	UI receipt and amount among job seekers; state education
CSPP party control; governor elections	State × year; election	Partisan DiD; close-election RD
Go-live dates (assembled)	17 + 9 states	**Treatment**: first and second wave

All data are public; the new input is a data set of system launch dates built from state audits, press releases, and vendor records.

Full inventory with variables and years in the paper's appendix. LAUS [BLS state unemployment rates]; CSPP [Correlates of State Policy, party control] (Grossmann et al., 2021).

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All Work-Search Law Checks



	Effect (SE)	States
All dated launches	0.740*** (0.273)	14
Drop MI, MS, MO [raised contacts near go-live]	0.936*** (0.307)	11
Drop all 7 states whose requirement changed	0.664*** (0.253)	7

No. Without the states that raised required employer contacts near launch the effect is larger (0.94), so it is not a change in the written rule.

Plot: event study without MI, MS, MO. Required employer contacts [number of employer contacts a claimant must report each week] from DOL's *Comparison of State UI Laws*, 2012–2020 editions, which cover every launch in the main sample. In the seven-state sample the pre-trend test rejects.

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Were the Extra Denials Correct? DOL Audits

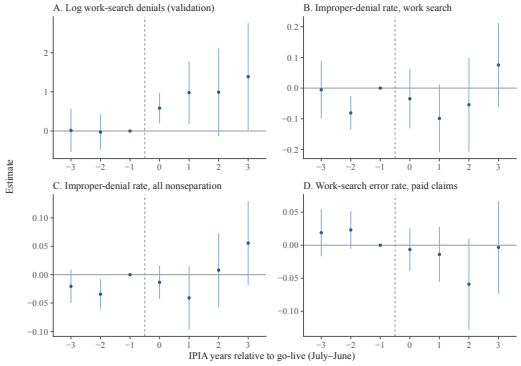
	Pre-launch mean	Post minus pre (SE)	Pre-trend *p*
Log work-search denials per 1,000 weeks (validation [same sample, confirms denials rose])	14.4	0.99*** (0.33)	0.84
Improper-denial rate, work search [share of audited denials auditors found wrong]	0.152	0.001 (0.058)	0.01
Improper-denial rate, all nonseparation denials	0.135	0.021 (0.015)	0.01
Work-search error rate, paid claims [share of audited paid weeks where the claimant had not met the search rule]	0.128	−0.035 (0.022)	0.17

Validation: denials rise by 0.99 in the audit sample. Legally improper denials do not rise; paid weeks without job search fall by 3.5 points, not significant ($p = 0.12$).

BAM [Benefit Accuracy Measurement: weekly random audits of paid and denied claims] (DOL, 2026a) by state and IPIA year [July–June], July 2011–June 2019 [no COVID months]; 9 states launching in this window, ±3 years. Including July 2019–June 2020: −0.046 (0.020). Auditors judge legal correctness, so a claimant who searched but did not document it counts as properly denied.

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Were the Extra Denials Correct? DOL Audits, Event Study

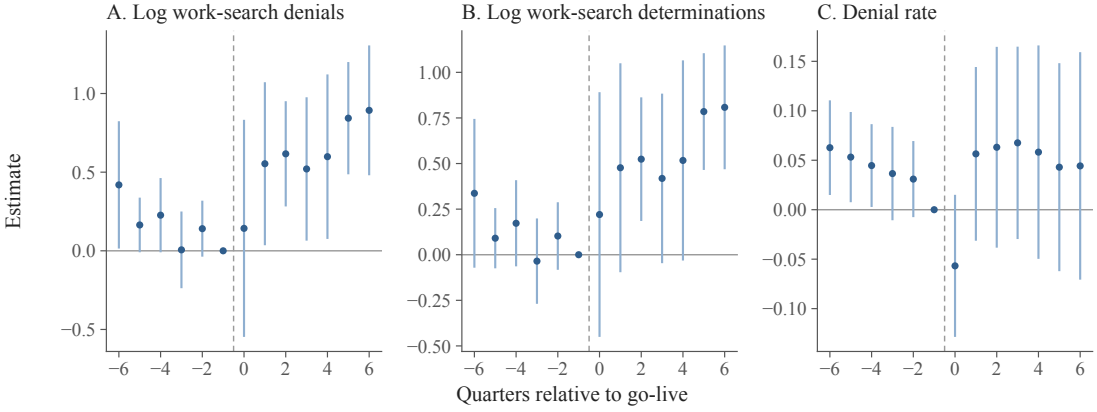


Improper-denial rates do not rise detectably; paid weeks without job search fall by 3.5 points, not significant ($p = 0.12$).

BAM [Benefit Accuracy Measurement: random audits of paid and denied claims] (DOL, 2026a), state × July–June year, July 2011–June 2019; 9 states; ±3 years.

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Second Wave: Denials, Determinations, Rate

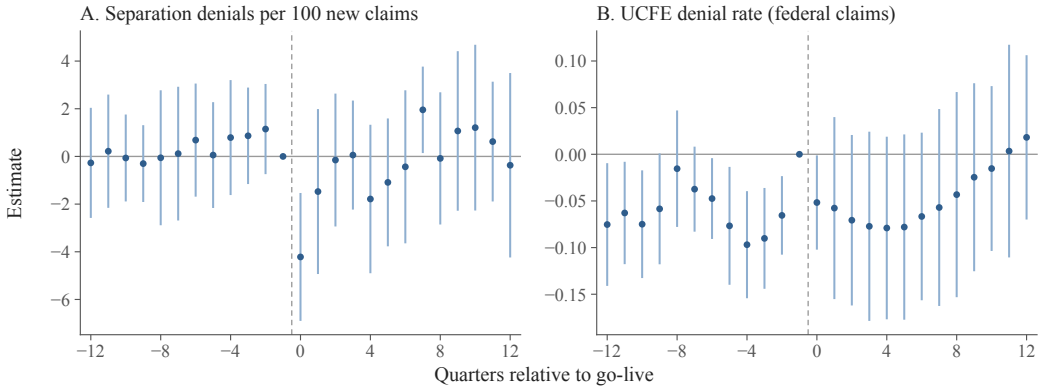


The second wave reproduces the margin: determinations rise (+0.54), the denial rate does not (+0.04, SE 0.05).

Stacked event studies, 2022Q1–2026Q2, nine launches 2023Q4–2025Q3, ±6 quarters; comparison states exclude the 2020–21 launchers.

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All Placebos



Separation denials do not move. The UCFE denial rate does not move but fails the pre-trend test; the UCFE determinations series (main slide) passes it.

Quarterly stacked event studies, 14 states, 2010–2019. Potential duration (legislated): −0.23 (0.38).

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Did Work-Search Rules Change within Two Years of Launch?

State	Go-live	Contacts/week (DOL editions)	Within ± 2 yrs
NM	2013Q1	2	No
MA	2013Q3	3	No
MI	2013Q4	none $\rightarrow$ 2 (2015)	Tightened
FL	2013Q4	5 $\rightarrow$ 3 rural (2015)	Loosened
IN	2014Q1	3	No
ID	2014Q3	1–3 $\rightarrow \geq$2 (2015)	Reclassified
TN	2016Q2	2 $\rightarrow$ 3 (2013)	No (3+ yrs before)
MS	2016Q2	none $\rightarrow$ 3 (2015)	Tightened
MO	2016Q4	1–3 $\rightarrow$ 2–3 (2015)	Tightened
WA	2017Q1	3	No
SC	2017Q3	4	No
ME	2017Q4	3 $\rightarrow$ 1 (2015)	No (3+ yrs before)
WY	2018Q2	2	No
NC	2018Q3	5 $\rightarrow$ 3 (2019)	Loosened

Sample	Effect (SE)	Pre p
All 14	0.739^{***} (0.273)	0.71
Drop 3 tightened	0.936^{***} (0.307)	0.11
Drop 5 with any change	0.956^{***} (0.363)	0.00

Drop 5: post$-$pre 1.02 (0.34). Second wave: no change within ± 2 years in editions through 2023 (KS, NV changed $\sim$4 years before).

8 of 14 states had no change within two years of launch; removing every state whose rule moved near launch, in either direction, makes the effect larger, not smaller.

DOL *Comparison of State UI Laws*, 2012–2023 editions (DOL, 2023); a change first shown in edition E is dated to $(E-1, E]$. 2015 edition redesigned the table; only FL verified against state rule. Full table for all 32 launches in the paper appendix.

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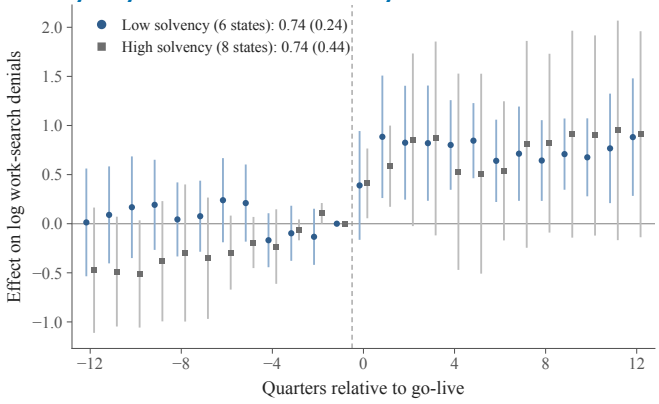
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Go-Live Event Study by 2007 Solvency

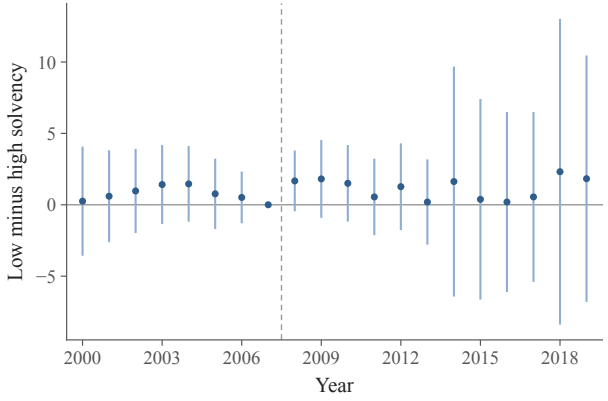


Both groups jump at go-live by about the same amount; the healthy-trust-fund states drift up before launch, the depleted ones do not.

Main stacked event study (log work-search denials, ±12 quarters, 2010Q1–2019Q4) run separately for low-solvency (IN, MA, MI, MO, NC, SC) and high-solvency (FL, ID, ME, MS, NM, TN, WA, WY) launch states; same never-modernized controls. 95% CIs, SE clustered by state.

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Denials in Low- vs. High-Solvency States, 2000–2019



After the recession, states with depleted trust funds did not raise work-search denials more than other states.

Difference in work-search denials per 1,000 weeks claimed, low minus high 2007 solvency, relative to 2007; 51 states; state and year FE, unemployment rate and its square; 95% CIs, SE clustered by state. Pre-2008 joint test $p = 0.46$.

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Decomposition: A Simple Numerical Example

$$\underbrace{\frac{D}{W}}_{\text{Denial rate}} = \underbrace{\frac{F}{W}}_{\text{Determination rate}} \times \underbrace{\frac{D}{F}}_{\text{Conditional denial share}}$$

	Before	After
Weeks claimed (W)	1,000	1,000
Formal determinations (F)	100	200
Denials (D)	20	40
Determinations per 1,000 weeks	100	200
Conditional denial share (D/F)	20%	20%
Denials per 1,000 weeks	**20**	**40**

Denials double because more issues are formally adjudicated, even though the conditional denial share stays at 20%.

Back to Results